

How to create a turbine blade using Fusion 360

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1 Before getting started

This is a step-by-step guide to implementing your blade geometry in the CAD program Fusion 360. The steps are based on version v2704.1.53, but the guidance is applicable to other versions.

Please be aware that:

- It is very important **to check that all dimensions are correct**. You will not have a second chance to print your blade, so make sure that the dimensions are correct before sending the file for printing. This guide will show you how to check the dimensions.
- The information on where and when to send the .STL file for printing can be found on Canvas. The material used is PLA, and the 3D printer is Neptune 4 Max.

2 How to get started

You can get a one-year student licence for Fusion 360 (and most other Autodesk products) [here](#). A screenshot of your student ID-app is sufficient for identification. Create an Autodesk account, choose Education license and download Fusion 360. The account will also be used to run Fusion 360. We recommend using your student e-mail address.

1. Start Fusion 360 and log in with your Autodesk Account.
2. Sometimes Fusion will not give you the Layout grid. In order to get the different axis, click on the “Grids and Snaps” icon and check the box called “Layout grid”. The grid and axis should now appear.

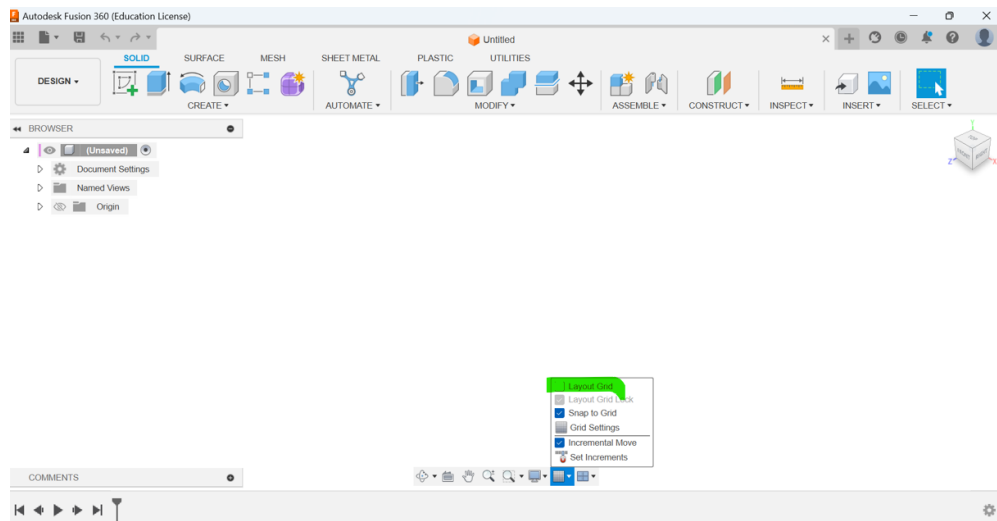


Figure 1

3. In order to create a new project, follow the Guide provided in the link: <https://www.autodesk.com/learn/ondemand/curated/fusion-quick-start-guide> You are free to call the project whatever you want and sort the files the way you prefer. We recommend organizing all the files in folders with accurate names to make it to easier sort through the files.
4. From the Figure below can we see that we have made a Project called “Design of a wind turbine” and created a new folder called “Test 1”. The file is still called “Untitled” since it is still unsaved. You can save a design file by clicking the save (disk) icon in the upper left corner of the toolbar (horizontally left from the name of the file “Untitled”). Or use the short cut CTRL +S (Windows) or CMD +S (Mac)

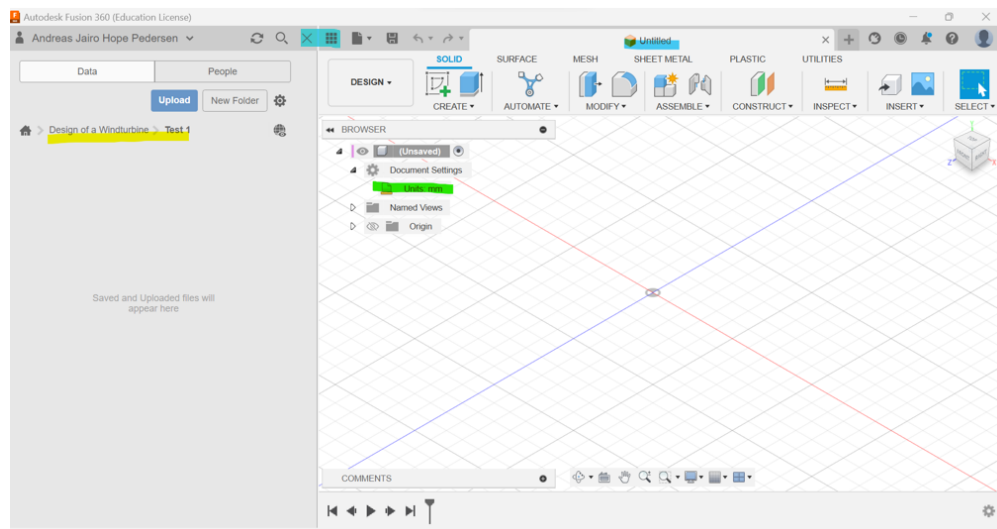
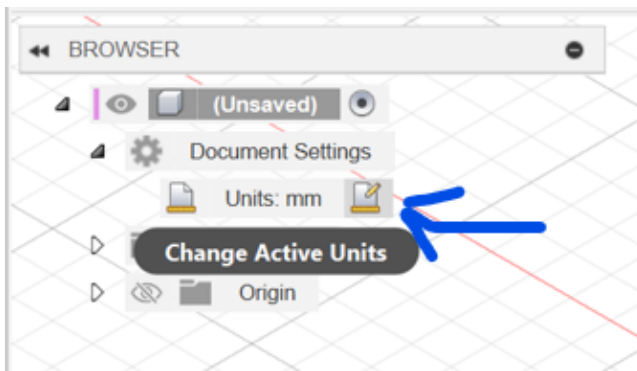
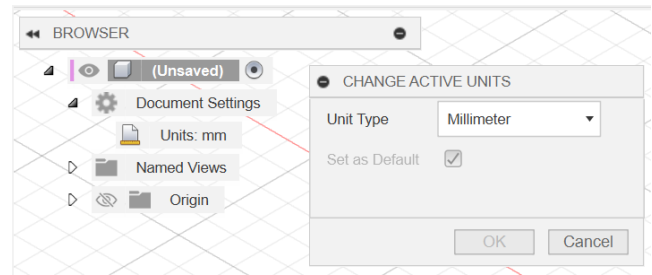


Figure 2

5. You can open and close the “Datapanel” by clicking on the 9 dots marked in blue in 2, or hide it by clicking the X marked in blue.
6. Finally check that your units are in millimeters (green marker). This can be done by clicking on “Document settings”, hover the mouse over “Units” and click on “Change Active Units”. The window on the right picture below should now appear and change the “Unit type” to “Millimeter”



(a)



(b)

Figure 3

3 How to Navigate in Fusion 360 and important key functions

- We would highly recommend using a PC-mouse and not your Touchpad. A lot of commands are much faster when using the PC-mouse and it will simplify a lot of the 3D modelling.
- For fast access to the different functions hit the S-key on your keyboard. You can simply start to search for a command. If you don't remember where a command is, no problem. This can help you quickly access every command across Fusion 360.

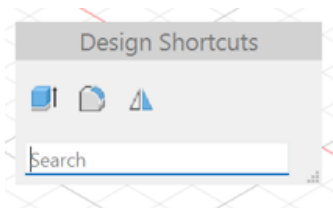


Figure 4

- In order to escape any command press “Esc” on your keyboard, e.g. if you have miss clicked “S” on your keyboard and do not want to use the “Design Short cuts” simply just press “Esc”.
- If you do not find your sketch, click the “Home” button in the upper right corner and Fusion 360 will orient your view to see your sketch(es) and/or model(s).

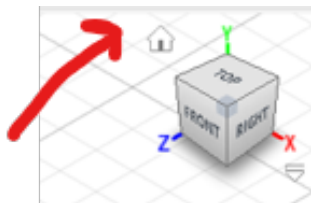


Figure 5

- The cube in 5 can be used to orient your view, try to play around with it and see that you can orient your view from the Front, Right, Top etc...
- Fast way to orient your design is by using the mouse, Zoom can also be accessed by scrolling the wheel on a mouse, and Free Orbit if you click and hold the scrolling wheel. Knowing these tips will make navigation much easier when you're drawing! see table below for an easy overview of the different hot keys.

- How to navigate with mouse and trackpad

Pan	press middle button	press middle button / two-finger drag
Zoom	roll wheel	roll wheel / pinch
Orbit	Shift + press middle button	Shift + press middle button / Shift + two finger drag

- You can also use “Free Orbit” from Figure 7 to freely rotate your design in any direction you want.

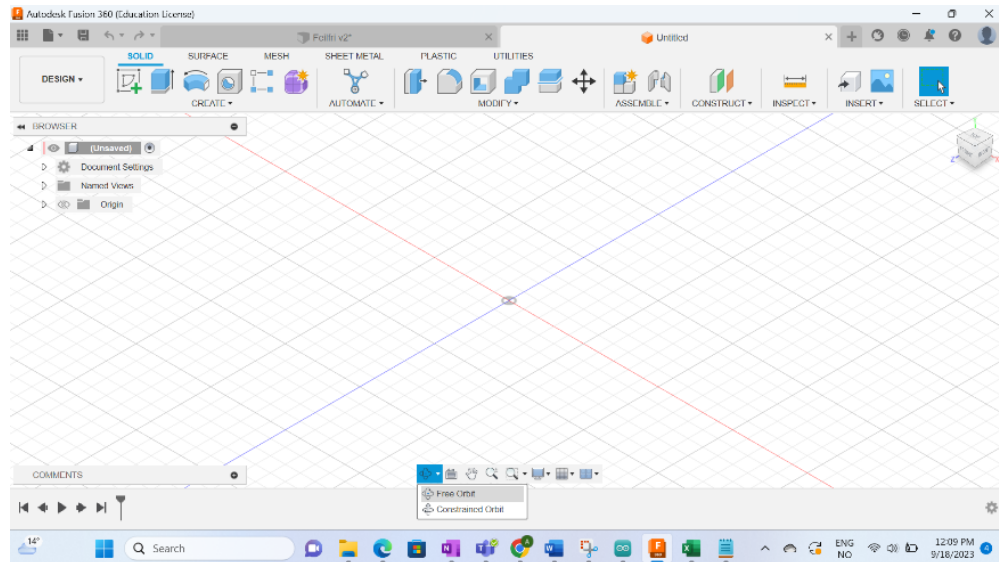


Figure 6

- Continuously save your file. For each time you save you create a new version. Old versions can be accessed if needed. The guide on how to do so can be seen in the link below, (this is not useful unless you need to access old versions). <https://help.autodesk.com/view/fusion360/ENU/?guid=ASM-OPEN-OLDER-VERSION>
- Use the timeline to move forward and backwards in your progress: <https://help.autodesk.com/view/fusion360/ENU/?guid=ASM-USE-TIMELINE>

4 Save your files as CSV files

7. Open Excel file 'Blade_coordinates.xlsx', sheet '1-Blade sections'. Copy paste your values of *local radius*, *chord length* and *twist angle* from Excel file attached to Assignment 4 'Adjust blade geometry_Fall26'

local radius [m]	chord length [m]	Twist [deg]
0.03	0.048	0
0.0675	0.21403845	30.037624
0.1125	0.1909117	20.386558
0.1575	0.15896106	14.489496
0.2025	0.13296593	10.682186
0.2475	0.11318682	8.0689358
0.2925	0.09807935	6.180417
0.3375	0.08631744	4.7582557
0.3825	0.07696416	3.6515215
0.4275	0.06937801	2.7671254
0.45	0.06937801	2.7671254

Figure 7

- Copy the relative coordinates (i.e. x/c and y/c) in the designated columns (cells J and K). Extend number of rows if necessary.

Instructions for the Users:

- Insert your values
- Copy paste your values of local radius, chord length and twist angle from Excel file attached to Assignment 4 'Adjust blade geometry_Fall26' (cells A to C)
- Copy the relative coordinates (i.e. x/c and y/c) in the designated columns (cells J and K)
- If the number of coordinates for your airfoil is more than the table marked below, extend the columns named 'Translated x/c ', 'Translated y/c ', 'Section 2' and so on upto the same number of rows

What's going on in the sheet?

- The cylinder/airfoil profile has been translated to locate the center of the airfoil chord at the origin (cells (J,K) and (O,P))
- Airfoil/cylinder has been scaled up by the factor of the chord length/diameter, transformed to [mm], and the coordinates are now rotated about the origin by the respective twist angle
- The coordinates are put together, along with the Z coordinate, which is the local radius, separated by commas (in next sheet "3-Data to be Imported to Fusion"). Those are the data to be imported

local radius [m]	chord length [m]	Twist [deg]
0.03	0.048	0
0.0675	0.21403845	30.037624
0.1125	0.1909117	20.386558
0.1575	0.15896106	14.489496
0.2025	0.13296593	10.682186
0.2475	0.11318682	8.0689358
0.2925	0.09807935	6.180417
0.3375	0.08631744	4.7582557
0.3825	0.07696416	3.6515215
0.4275	0.06937801	2.7671254
0.45	0.06937801	2.7671254

Section 1: cylinder			
x/c	y/c	Translated x/c	Translated y/c
1.0000	0.0000	0.49939	-0.01744
0.9908	0.0954	0.4902	0.07796
0.9636	0.1873	0.46298	0.16986
0.9193	0.2723	0.41873	0.25488
0.8597	0.3473	0.35906	0.32989
0.7868	0.4096	0.28618	0.39214
0.7034	0.4568	0.20276	0.43933
0.6125	0.4872	0.11187	0.46975
0.5175	0.4997	0.01684	0.48226
0.4218	0.4938	-0.07883	0.4764
0.3290	0.4699	-0.17162	0.45241

Other sections: airfoil			
x/c	y/c	Translated x/c	Translated y/c
1	0.0016	0.5	0.0008
0.95	0.0112	0.45	0.0104
0.9	0.0204	0.4	0.0196
0.8	0.0373	0.3	0.0365
0.7	0.0525	0.2	0.0517
0.6	0.0661	0.1	0.0653
0.5	0.0774	0	0.0766
0.4	0.0859	-0.1	0.0851
0.3	0.0905	-0.2	0.0897
0.25	0.0908	-0.25	0.09
0.2	0.0892	-0.3	0.0884

Figure 8

- From the sheet '2-Data to be Imported in Fusion' you should now have now gotten your final blade design 9.

	Section 1	Section 2	Section 3	Section 4	Section 5	Section 6	Section 7	Section 8	Section 9	Section 10	Section 11	Section 12
23,97072,-0,83712,30	92,56049681	89,42364931	76,92071652	65,31112943	56,02041994	48,74620017	43,00424793	38,40003473	34,64587819	34,64587819	0,0,0	0,0,0
23,5296,3,74208,30	82,26732298	79,83752083	68,84364749	58,54143633	50,26458786	43,76936729	38,63451252	34,51258291	31,14886877	31,14886877	0,0,0	0,0,0
22,22304,8,15328,30	72,01700554	70,27799415	60,78248743	51,78160191	44,51511074	38,79675809	34,26764119	30,62709176	27,65319907	27,65319907	0,0,0	0,0,0
20,09904,12,23424,30	51,67708202	51,25869753	44,71982601	38,29890313	33,03998762	28,86737845	25,54463882	22,86346201	20,66688370	20,66688370	0,0,0	0,0,0
17,23488,15,83472,30	31,51929807	32,35245857	28,72477777	24,85810376	21,59187308	18,95594941	16,83380876	15,10816512	13,68626220	13,68626220	0,0,0	0,0,0
13,73664,18,82272,30	11,53293960	13,55262680	12,79336545	11,45673913	10,16917839	9,061415066	8,134435002	7,360710940	6,710999647	6,710999647	0,0,0	0,0,0
9,73248,21,08784,30	-8,20699474	-5,09424461	-3,04657020	-1,88793806	-1,21697525	-0,80883316	-0,54847032	-0,37546935	-0,25655942	-0,25655942	0,0,0	0,0,0
5,36976,22,548,30	-27,6469345	-23,5549034	-18,7751429	-15,1636044	-12,5586441	-10,6495157	-9,21132713	-8,09792493	-7,21474034	-7,21474034	0,0,0	0,0,0
0,808319999999998,23,14848,30	-46,6690246	-41,7561947	-34,3486031	-28,3431486	-23,8383522	-20,4490174	-17,8462592	-15,8012639	-14,1598588	-14,1598588	0,0,0	0,0,0
-3,78384,22,8672,30	-55,9657878	-50,7238313	-42,0557883	-34,8836273	-29,4464314	-25,3276499	-22,1494048	-19,6431300	-17,6257194	-17,6257194	0,0,0	0,0,0
-8,23776,21,71568,30	-65,0589833	-59,5651094	-49,6874058	-41,3772773	-35,0243245	-30,1862200	-26,4389461	-23,4756829	-21,0852162	-21,0852162	0,0,0	0,0,0
-12,39024,19,73472,30	-73,8950406	-68,2467767	-57,2235695	-47,8117751	-40,5640879	-35,0194480	-30,7113030	-27,2964717	-24,5366746	-24,5366746	0,0,0	0,0,0
-16,169864,16,00874,30	-87,3168714	-76,6007741	-64,5688752	-54,1706877	-46,0775617	-39,8010070	-34,0407014	-31,0073751	-27,0705637	-27,0705637	0,0,0	0,0,0

Figure 9

- Now copy all the coordinates from the excel sheet. REMARK: You only copy the coordinates from ONE section at the time! Start with section 2, section 3, ... (The cylinder or section 1 will be created later in Fusion)
- Important:** make sure the decimal notation are “.” and not “,”. Click “File” à “Options” (bottom left) à “Advanced” Uncheck “Use system separators” and change both “Decimal separator” and “Thousands separator” to “.”.
- Now open the Notepad application shown in 10.

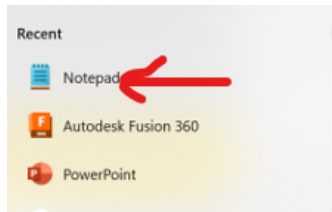


Figure 10

- And further paste in the coordinates copied from the Excel sheet. The coordinate-file should look like the file in Figure 11. In this example we have copied all the coordinates from section 2.

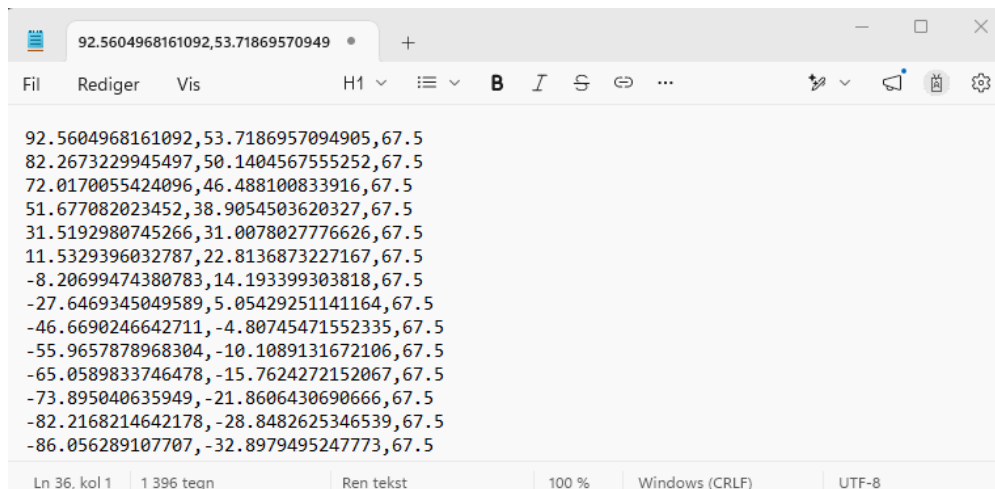


Figure 11

- Click on “File” à “Save As”. REMARK: When giving the file a name ends the name with .csv. Save file type should be “All files”

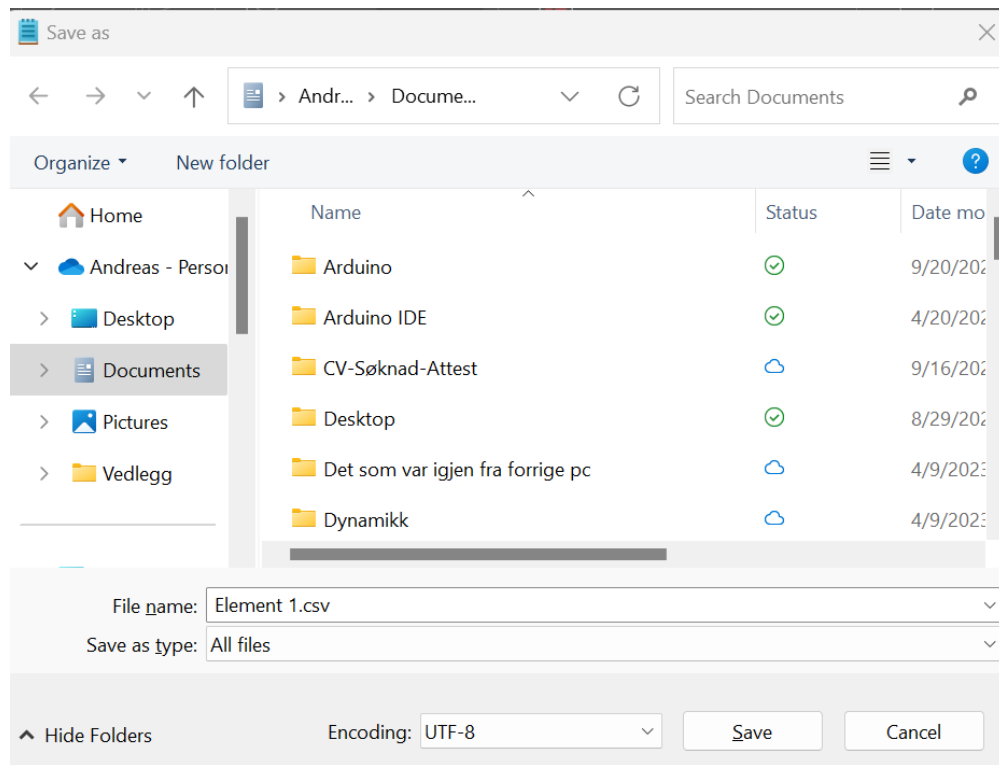


Figure 12

- Repeat this for all your blade elements, remember to give them all different names, we will suggest calling them “Element 1.csv”, “Element 2.csv” etc... In order to keep a good structure.

5 Importing and modifying the blade elements

- We are now back in Fusion 360. Start by saving your file in Fusion 360. All your blade elements should now be saved as separate .csv files in a folder on your computer.
- For the purpose of importing the .csv file, go to “UTILITIES” on the toolbar. Select “ADD-INS” à “Scripts and Add-Ins...”.

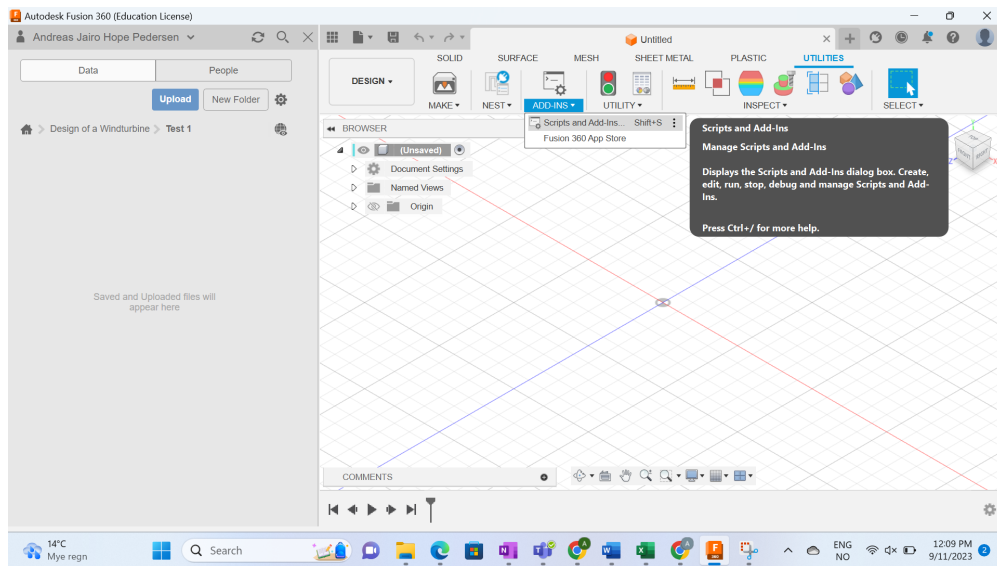
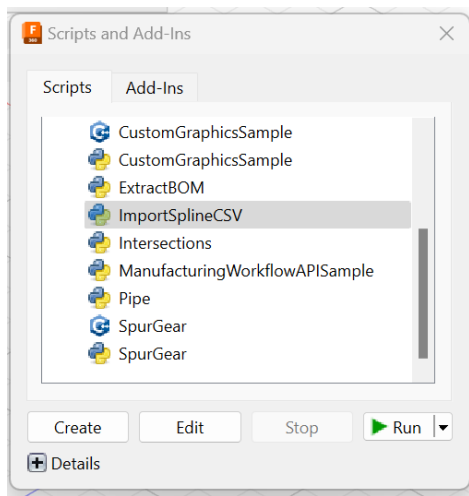
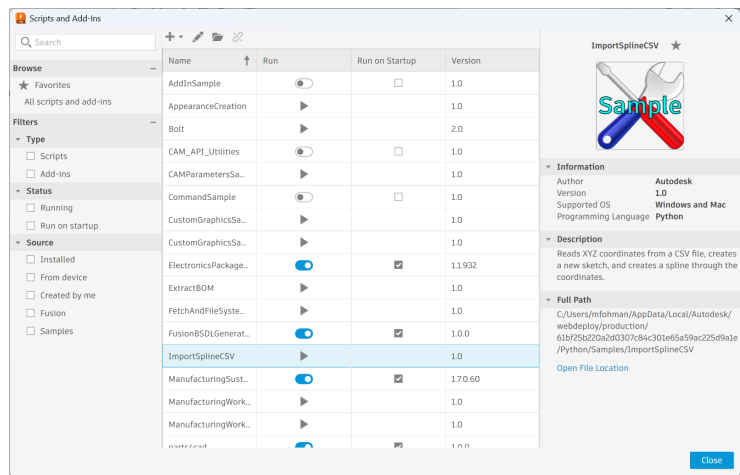


Figure 13

17. A new small window as seen in 14 should now appear called “Scripts and Add-Ins”, scroll down to “ImportSplineCSV”. Double-click to run the script and select the CSV file you want to Import. Note this window might look different depending on version.



(a) Alternative A



(b) Alternative B

Figure 14

18. All the elements will now appear in the folder you saved them in so remember which folder you save the .csv files in. As shown in 15 all the elements are there. Then you click on the element you want to import and click on “Open”.

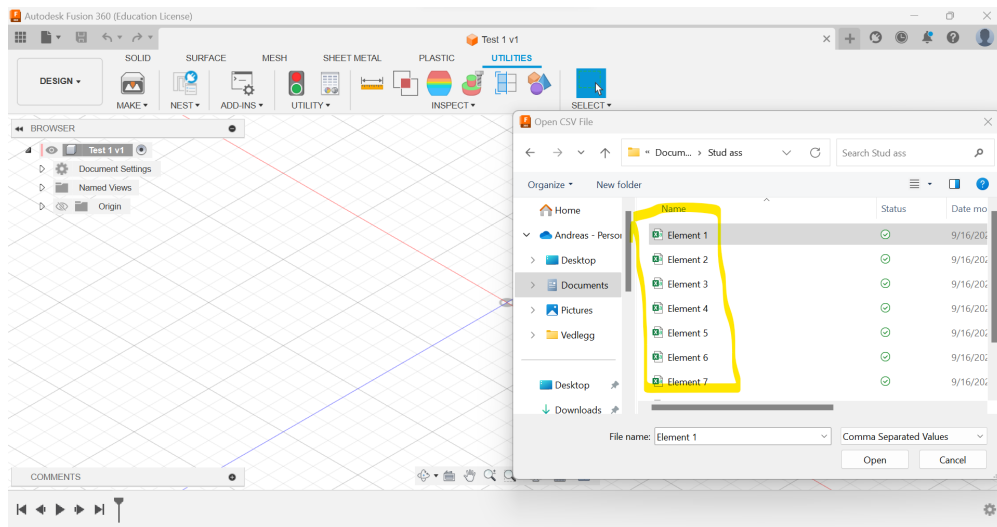


Figure 15

19. Your element should now appear. Here we show element 2 as an example. Orient it so you can see the sketch, either by using “Free orbit” or the little box in the top right corner and turn it to the “Front view”. You can now see that the centre of the blade is aligned with the centre of the two axes. If you do not find your sketch, click on the HOME icon.

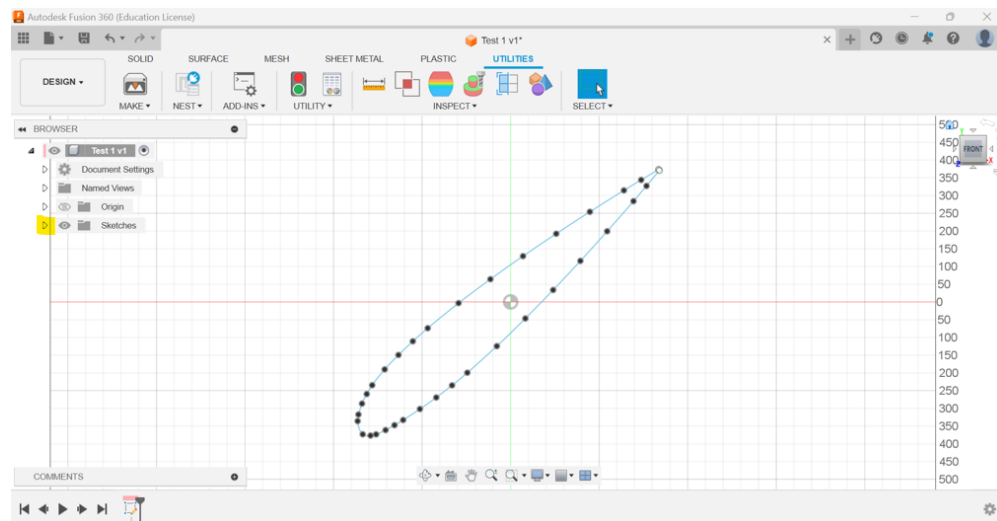


Figure 16

20. Click on the eyeball icon next to the sketch to show or hide it. Sketch visibility may be automatically turned off when performing a modelling action. You can see all your sketches by clicking on the little triangle marked in yellow in 16.
21. Now go back to SOLID on the toolbar. Then click on CONSTRUCT and choose “Offset plane”.

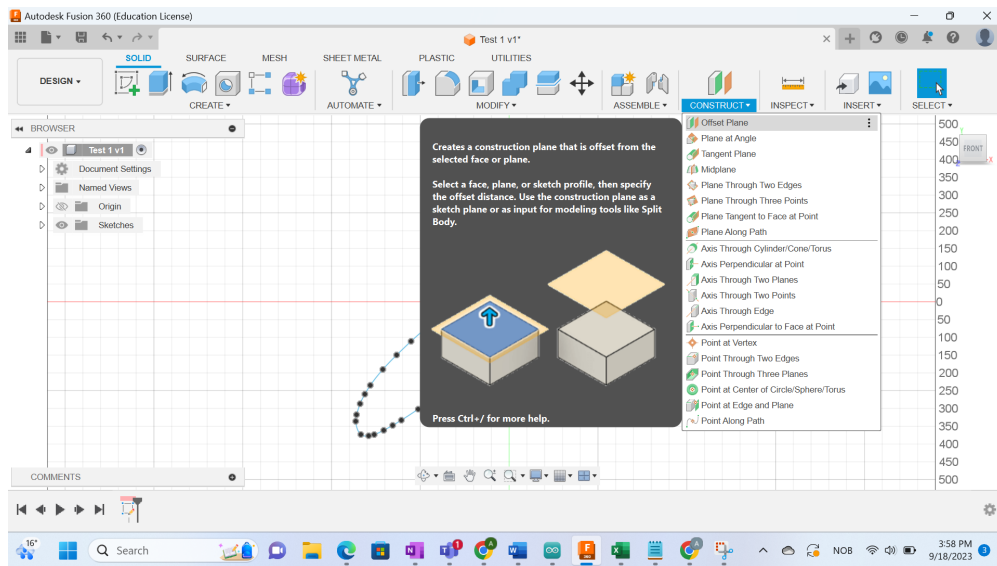


Figure 17

22. Choose the plane in the background and write in the distance. The distance is in the .CSV you made for that particular element; you can see in this example that the distance is the same as in 11, but you need to scale it up by 10. This is because Fusion 360 scales up the drawing by a factor of 10 when you import a CSV file. We will correct this at the end. The information about the distance can always be found in the CSV file in the last column.

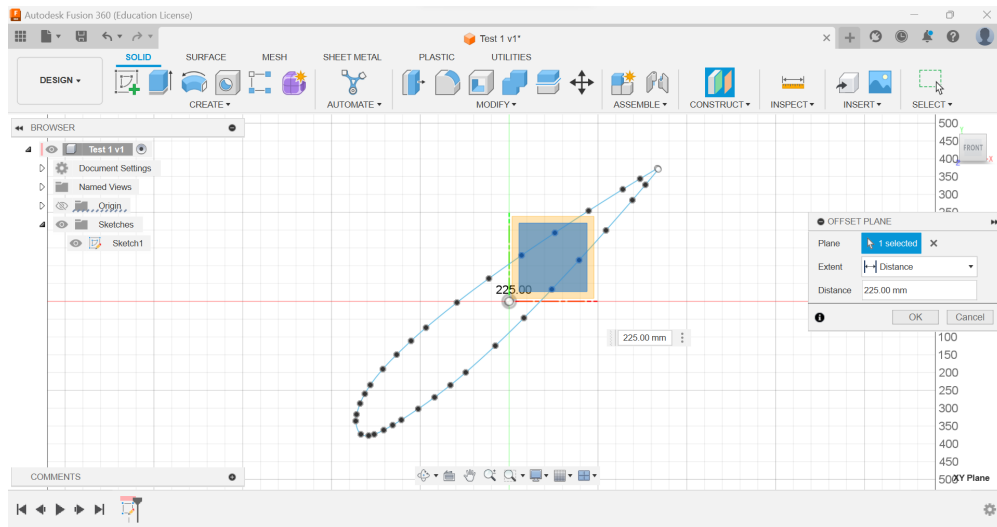


Figure 18

23. Click “OK”. Now use “Free orbit” to ensure that the drawing is in the same plane as the newly constructed plane. The plane can also be seen under “Construction”.

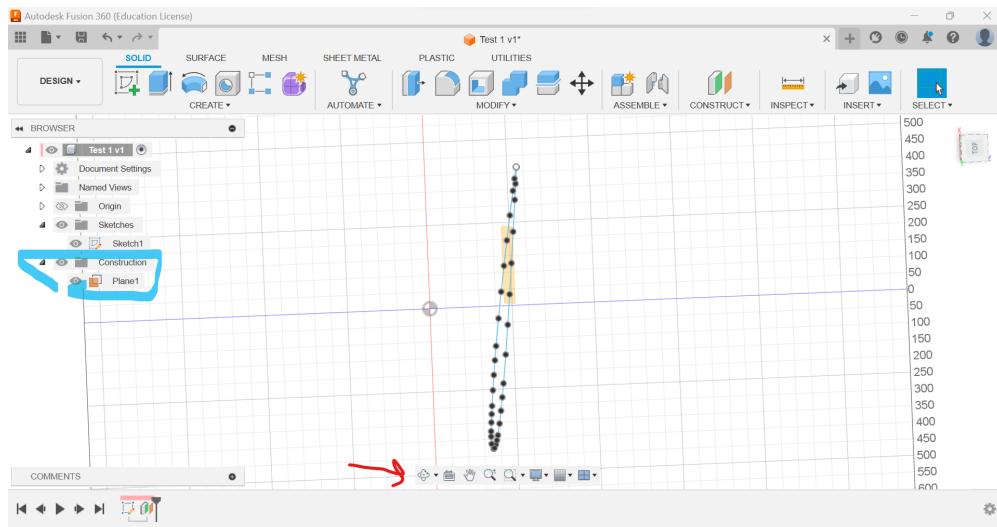


Figure 19

24. Now under SOLIDS go to “CREATE” and choose “Create Sketch”. And then choose the plane that you created.
25. The drawing will now orient itself into the newly created plane. Again, go to CREATE, and then choose “Project/ Include” à “Project” as shown in 21. OR just click “P” on your keyboard.

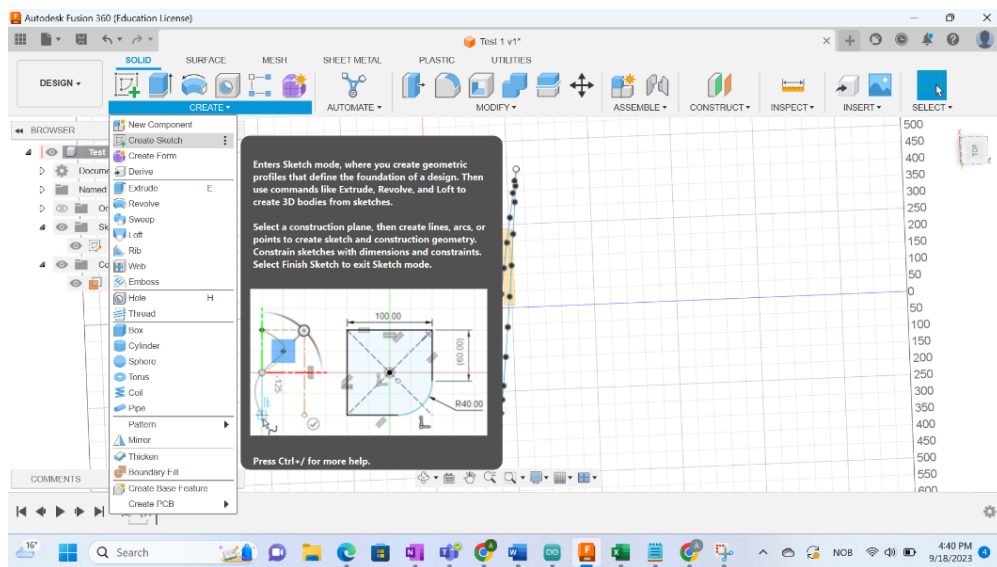


Figure 20

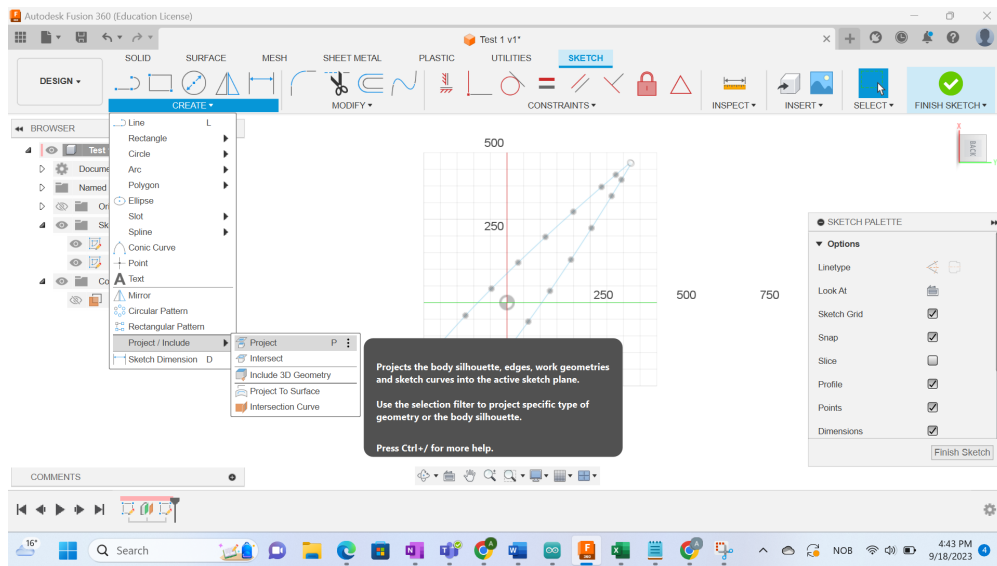


Figure 21

26. The little window called PROJECT should now appear. Also, the SKETCH PALETTE should appear. Make sure that 3D Sketch box is checked. When clicking on the element make sure that you click on the sketch itself as shown in figure 22a, and NOT on a point on the sketch as shown in 22b. If you click on a point by accident, just click “Esc” and try again. This can also be avoided by hiding Sketch 1 (Click on the eye icon to hide it, see the red arrow in Figure 24 showing that the eyeball is not checked/grey)

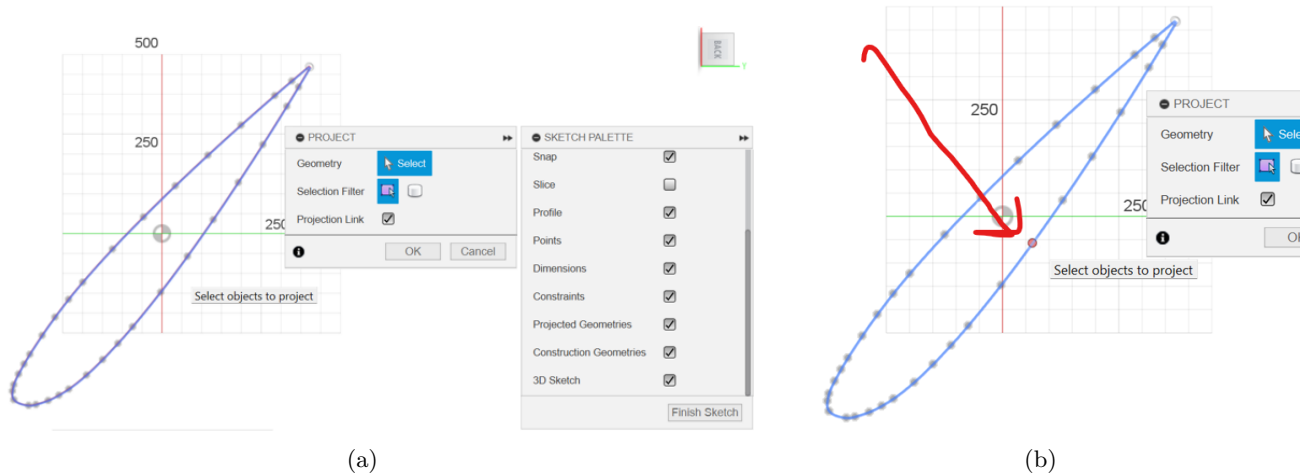


Figure 22

27. Now go to CREATE click on “Line” or click “L” on your keyboard. You will now be able to make lines and close the blade element. Hover over the sketch semi-close to the edge of the blade. When it goes from a black plus sign to a blue X it means that it will be constrained to the drawing of your element and that is when you left click.

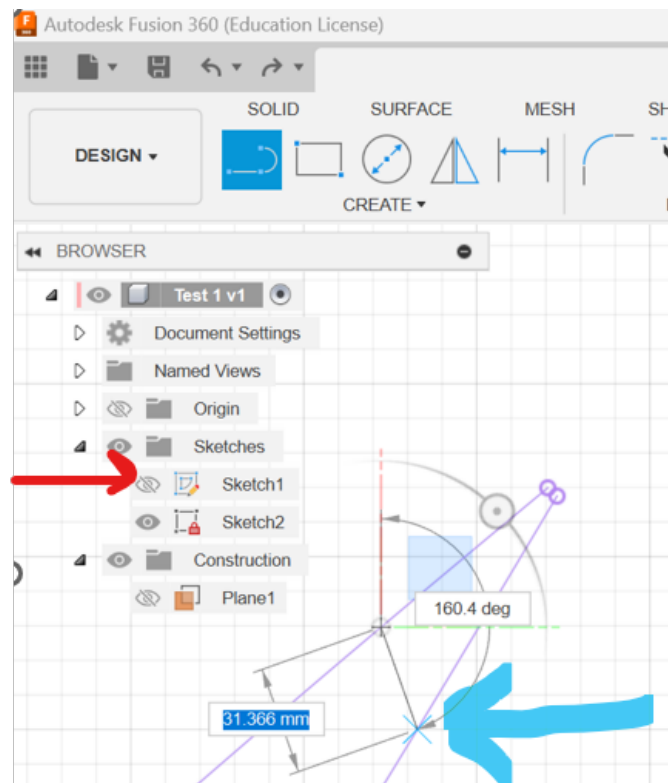


Figure 23

28. After you have created the line, the blade will look like the one in 24, and you can see that you closed the sketch since it turned light blue.

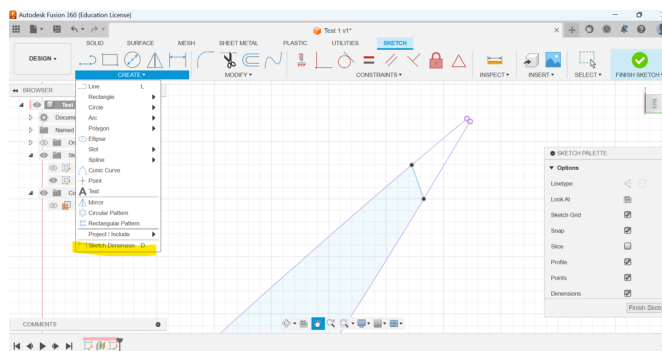


Figure 24

29. Now you will set the length of the line. This will be the thickness of the blade at the edge. This must be 6 mm or larger so that the blade does not break during machining later. (The entire blade will be scaled down by 10 in the end so in the end result it will be 0.6mm which is the limit for the CNC machine milling the blades). Also remark that for some of the larger elements it has to be larger in order to close the sketch, this error is more common close to the hub.
30. In order to set the length of the line, go to CREATE and choose "Sketch Dimensions" and then click on the blue line (25) or just click on the line and then click "D" on your keyboard. You might have to move your mouse in order to get the dimensions to show at first and then left click to have it show permanently. Double click on the dimension and change it to 6mm.

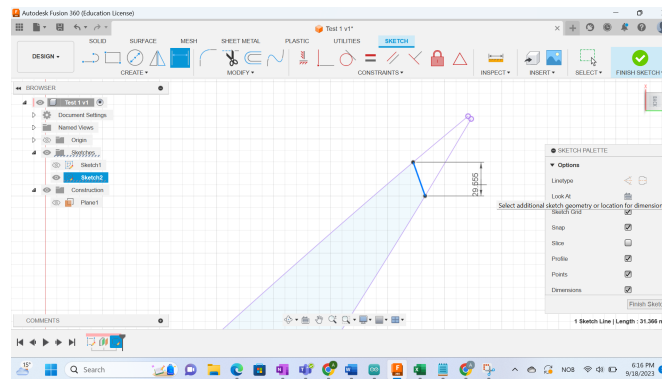


Figure 25

31. Now go to CONSTRAINTS and choose “Horizontal/Vertical” and then click on your blue line. If your sketch is still not closed (blue) you need to increase the length of the line as shown in 26.

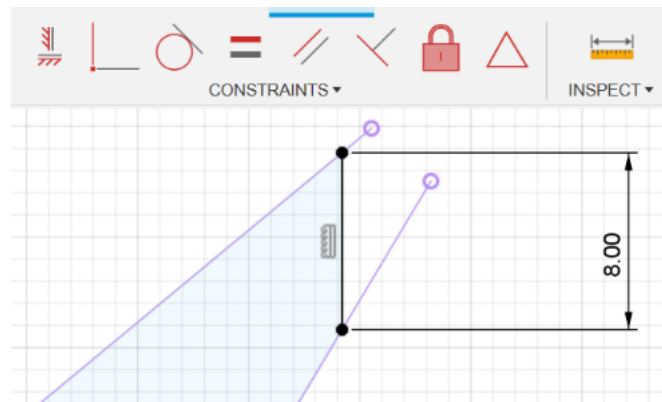


Figure 26

32. Before finishing up your sketch go to the BROWSER on the left and hover over your sketch. Then click on it, wait a second and then click again, you should now be able to rename your sketch. We would highly recommend naming your sketch the same as you called it when it was a CSV-file.

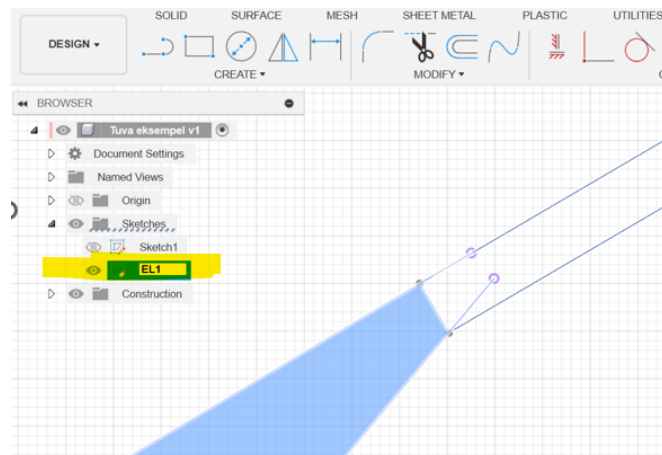


Figure 27

Important note: When you close the sketch, make sure that the thickness at the end of the element is at least

6 mm. 27 is just one way of doing it however a line perpendicular to the edges of the element leading to blunt edges is recommended for better finish.

33. Then you can click FINISH SKETCH. Make sure that you always hide the CSV sketch you got first so it doesn't confuse you in your future work. We would also recommend hiding all sketches and old planes when you repeat this method for remaining elements. Now repeat step 15-33.
34. Just to show how this will look like an example with three elements is shown in 28, where you can see that only element 3 is shown, all elements are given names and all other planes and sketches are hidden.

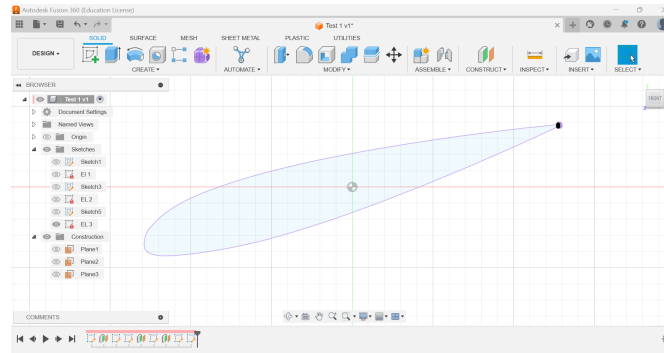


Figure 28

6 Creating the HUB-connection

IMPORTANT NOTE: your blade length should never be longer than 0.45 meters, INCLUDING THE HUB. In other words, your entire turbine blade should not span longer than 0.9 meters in total.

36. After all the elements are created you need to make the connection to the hub (circle). Create a offset plane at height 300 mm above background (30 mm * scale-up-factor). Go to CREATE, “Create Sketch” and choose the new offset plane. Rename the newest sketch to “Hub”.
37. Go to CREATE again and choose “Circle” à “Center Diameter Circle”. Centre the circle in origin as shown in 29 and make the diameter 480mm. Then click FINISH SKETCH.

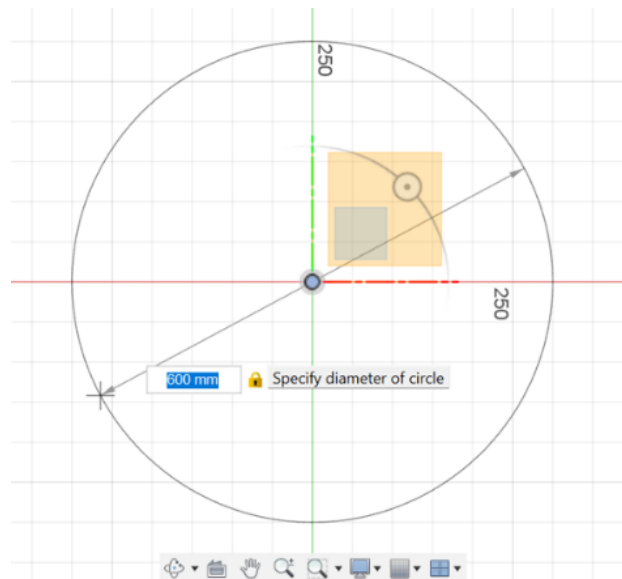


Figure 29

38. Now show all the sketches, (all the elements + the hub). If you click the Home button (the little house) in the upper right corner you can see all the elements and the hub connection.
39. Under SOLID go to CREATE and choose “Loft”. Start by clicking on the Hub-connection sketch and then click on each sketch in order to connect the elements together in order to create a new body as shown in 30.

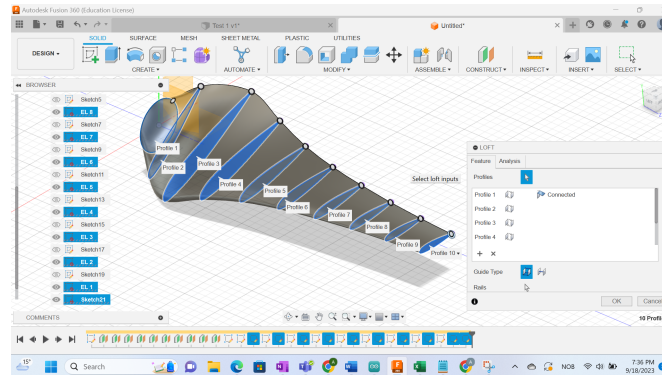


Figure 30

Common mistake: When lofting the elements together make sure that you loft each sketch and not from point to point. An example of lofting into a singular point is shown below where you get the white dot in the middle of the sketch. You do NOT want to do this mistake.

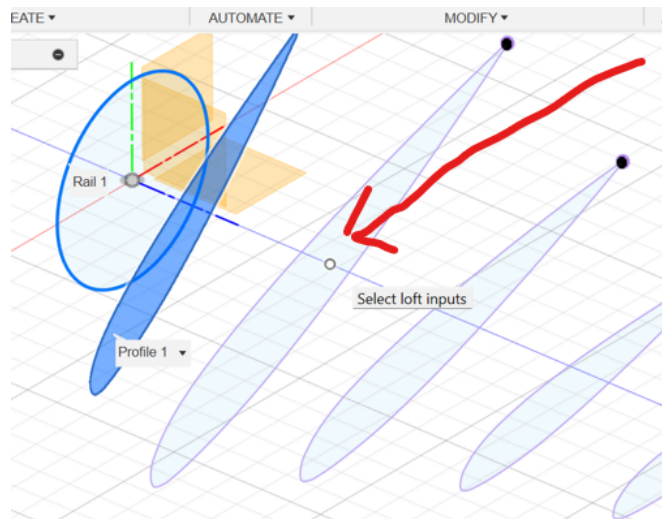


Figure 31

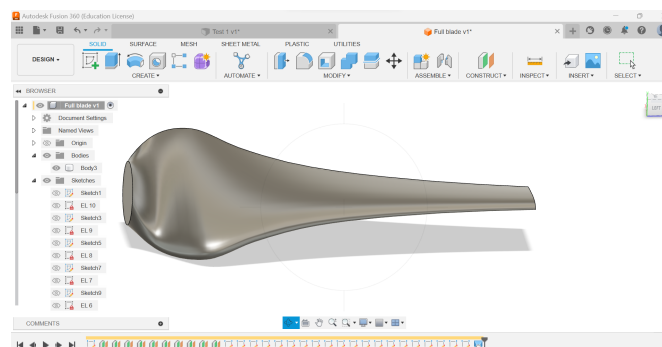


Figure 32

40. You will now have a complete body like the one in 33 blade 1. In order to scale it down, click on the body and click “S” to get the “Design Shortcuts” up. Search for “Scale” and chose the one with the blue icon.
41. Change the scaling from 1.00 to 0.1. Meaning that we are scaling it down to the right size.

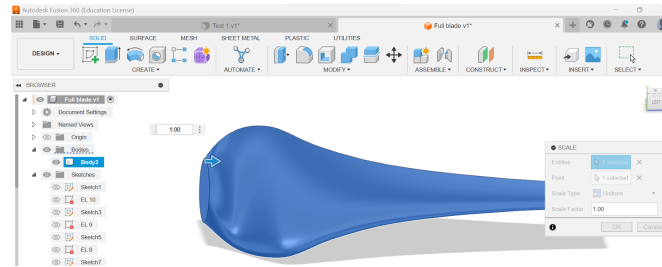


Figure 33

42. You might lose the blade but just click the “Home” icon to see your body. Now go to your toolbar and go to INSPECT à “Measurement”. First check that the large circle, the connection to the hub is 48 mm in diameter. And then check that the length of the blade matches your dimensions from QBlade and excel file.

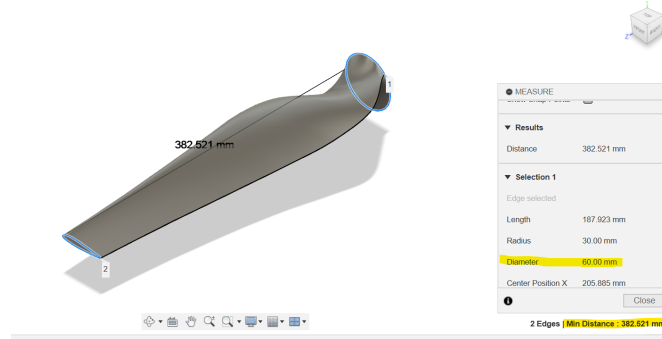


Figure 34

43. Save your design and get ready for the Hub.

7 The Hub

45. The hub will be given to you on Canvas either as a .STEP, .f3d or STL file. The file used in the assembly examples is just to show how to assemble. The final design might vary from the examples below.

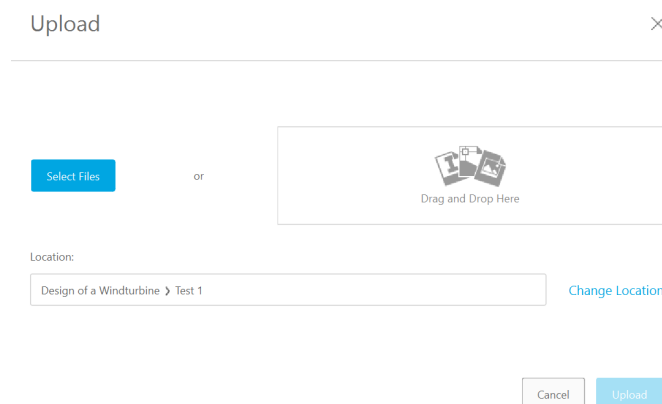


Figure 35

46. In order to upload the file into Fusion 360 open the data panel and click on “Upload”.
47. Further click on “Select Files”, the blue box below.
48. Go to downloads or the map you saved the Hub file in and choose your file and click “Open”. And then click “Upload”. Your file should now be visible in your folder in Fusion 360.

8 Assemble the parts

49. We will now assemble the parts for 3d-printing.
50. Create a assembly-design in the same folder as the Hub and Blade. Go to ”File”, ”New...”, and click ”Assembly Design”. Call the new design “Assembly” and save it. If you now “Show the data panel” as shown below:

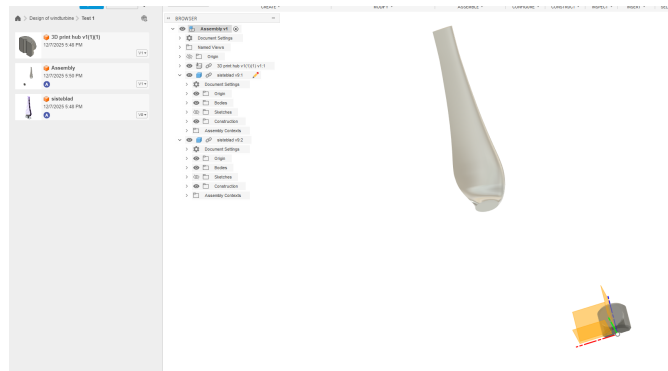


Figure 36

51. Now drag and drop the hub and the blade into the “Assembly” file as shown above in 36.
52. Under RELATIONSHIPS click on “Joint” marked below in 37.

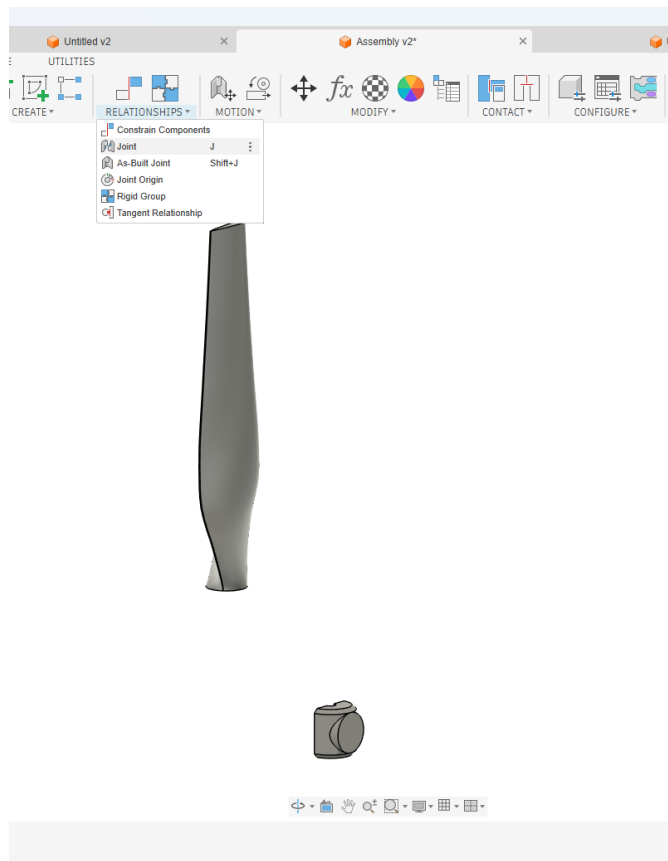


Figure 37

53. First choose one of the blades as shown below, and make sure that you click when the mouse is centred in the middle of the circle.

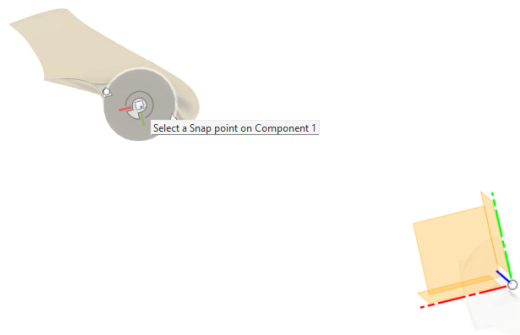


Figure 38: Enter Caption

54. Then pan the view over to the hub and click on one of the sides with no holes. Make sure that the joint also is centred in the middle here too.

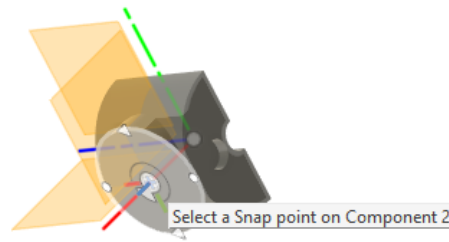


Figure 39: Enter Caption

55. The blade should now connect to the blade with a "rigid joint". The angle can be adjusted if needed, just orient the view from the side to see that you have the right twist angle on the blade. Your blade should rotate counter clockwise. The orientation related to the wind is shown in 40.

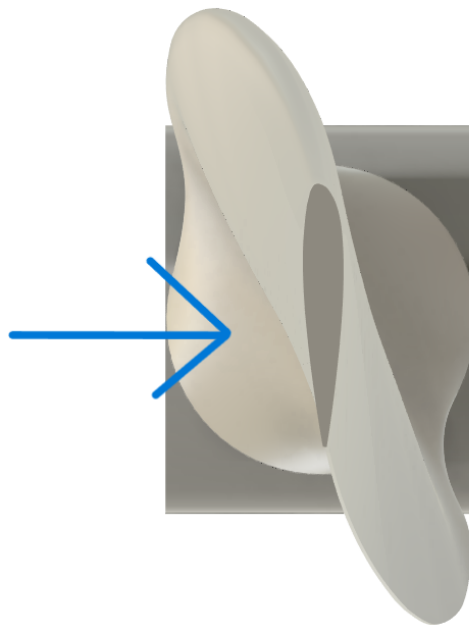


Figure 40: Blade orientation with wind direction. The suction side is on the right, with the pressure side on the left.

56. Now we are almost done! Remember to check the blade angle, and that the body includes both blade and hub. Check this with the teaching assistants before going any further!
57. Below is a picture of a finished blade. Remember that the blade is supposed to rotate **anti-clockwise**, so remember that the leading edge should point upwards, with the suction side at the back of the turbine. Remember that the full length of your **blade does not exceed 0.45m** including the hub. Your blade should now look something like this:



Figure 41: Full assembly for 3d-printing.

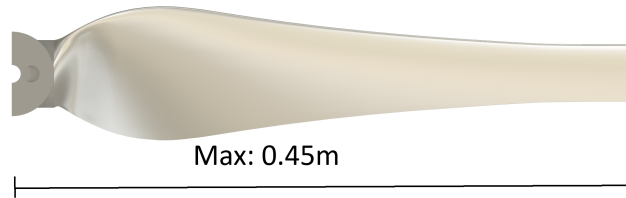


Figure 42: Maximum length of 0.45m

58. In order to export your file as a STL file right-click on the Assembly, and click "Save as mesh".

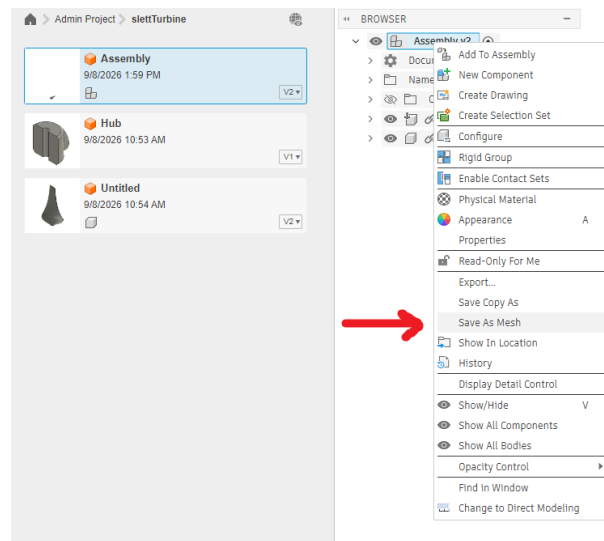


Figure 43

59. Choose STL Binary and Refinement = "high".

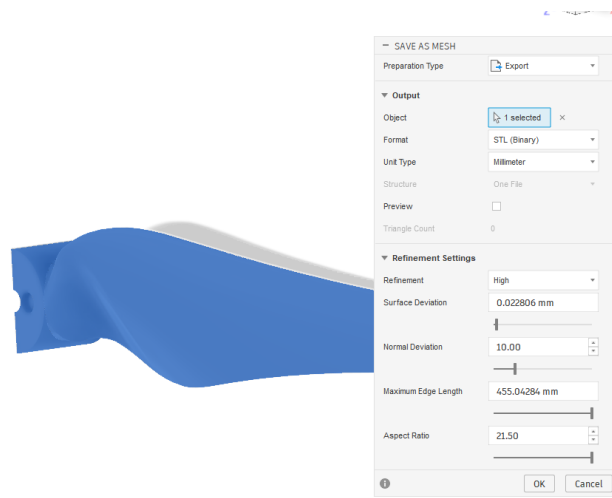


Figure 44

60. Press Ok and save.

61. EXTRA NOTES: Make sure that the twist angle is correct; if you do not use the twist angle from the original blade design (in Python or QBlade), your entire wind turbine can fail! If the angle is too low, the parts of the wings close to the hub and near the end, will counter each other! Entire projects have failed during testing in wind tunnel because of minor errors in correct blade angle. Also, always remember to stay critical; “Does this construction work? Is the blade properly angled? Is it inside the member dimensions?”